

An
Industry Oriented Mini Project Report
on
**ONLINE FOOD ORDERING SYSTEM UNDER
E-COMMERCE**

Submitted in partial fulfillment of the requirements for the award of degree

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the project report titled **“ONLINE FOOD ORDERING SYSTEM UNDER E-COMMERCE”** is Being submitted by **B.Pavan(237Z1A0511),J.Charan(237Z1A0563) and B.Nithil(237Z1A0525)** in Partial fulfillment for the award of **Bachelor of Technology in Computer Science and Engineering** is a record bonafide work carried out by them. The results embodied in this report have not been submitted to any other University for the award of any degree.

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DECLARATION

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ABSTRACT

The rapid growth of internet and digital technologies has significantly influenced modern business practices, especially in the field of electronic commerce. This mini project focuses on the design and development of an Online Ordering System under an E- Commerce Website that enables customers to conveniently browse products, place orders, and complete purchases through an online platform. The system provides core functionalities such as user registration and authentication, product catalog display, shopping cart management, order processing, and secure online payment. An administrative module is also included to manage product details, monitor customer orders, and update order status efficiently. By automating the ordering process, the system reduces manual effort, minimizes errors, and improves overall operational efficiency. The user-friendly interface enhances customer experience and accessibility, while real-time order management ensures timely and accurate service. This project demonstrates the practical application of e-commerce concepts and offers a scalable solution suitable for small to medium-sized online businesses in today's competitive digital marketplace.

Keywords—•Admin Management System, User Authentication, Product Catalog Shopping Cart, Order Processing, Secure Payment Gateway, E-Commerce Platform.

TABLE OF CONTENTS

	Page No.
Abstract	
List of Figures	i
List of Tables	ii
List of Abbreviations	iii
1. INTRODUCTION	1
1.1 Motivation	1
1.2 Problem Definition	2
1.3 Objective of project	2
1.4 Limitation of project	2
2. LITERATURE SURVEY	4
2.1 Introduction	4
2.2 Existing System	5
2.3 Proposed system	6
3. SYSTEM ANALYSIS	7
3.1 Functional reqyirements	7
3.2 Non-Functional requirements	8
3.3 Software requirements	9
3.4 Hardware requirements	9
4. SYSTEM DESIGN	10
4.1 Introduction	10
4.2 UML Diagrams	10
4.3 Modules	16

5. IMPLEMENTATION & RESULTS	17
5.1 Introduction	17
5.2 Method of Implementation	17
5.3 Algorithm and Flowcharts	25
5.4 Control Flow of the implementation	25
5.5 Output Screens	26
6. TESTING	29
6.1 Introduction to Testing	29
6.2 Types of Tests considered	29
6.3 Various Test Case Scenarios Considered	30
7 CONCLUSION AND FUTURE ENHANCEMENT	31
7.1 Project Conclusion	31
7.2 Future Enhancement	31
8 REFERENCES	32
8.1 Journals	32
8.2 Books	32
8.3 Sites	32

LIST OF FIGURES

S. No.	Figure No.	Name of the Figure	Page No
1	2.2	Existing System	5
2	3.5	Block diagram of system	8
3	4.2.1	Class diagram	11
4	4.2.2	Activity diagram	13
5	4.2.3	Use case diagram	14
6	4.2.4	Sequence diagram	15
7	5.5.1	Output Screen QuickBite Home page	26
8	5.5.2	Output screen Feature Selection of QuickBite	27
9	5.5.3	Output screen AI Smart Meal Planner	27
10	5.5.4	Output screen Menu Browsing Section	28
11	5.5.5	Output screen Food Item Detail Page	28

LIST OF TABLES

S. No	Table No	Name of the Table	Page No.
1.	6.3	Test case for proposed system	30

LIST OF ABBREVIATIONS

S.NO	Abbrevation	Defination
1	OFOS	Online Food Ordering System
2	OFDS	Online Food Delivery System
3	E-FOS	Ecommerce Food Ordering System
4	FOMS	Food ordering Management System
5	OFOA	Online Food Ordering Application
6	IFOS	Internet based Food Ordering System

1. INTRODUCTION

The Online Food Ordering System is an e-commerce application that allows customers to order food from restaurants through the internet. It provides a convenient platform where users can browse menus, select food items, and place orders online. This system helps customers save time by avoiding long waiting lines at restaurants. It also allows restaurants to manage orders efficiently and improve customer service. Users can view different restaurants, compare menus, and choose their preferred meals easily. The system typically includes features such as user registration, login, menu display, cart management, and order tracking. Online food ordering systems are widely used through websites and mobile applications. They help restaurants increase their sales and reach more customers. Secure payment methods and quick delivery options make the process simple and reliable. Overall, the system enhances the food ordering experience through digital technology and e-commerce services.

1.1 Motivation

The development of an Online Food Ordering System is driven by the need to make food ordering more convenient and efficient. In modern society, many people have busy schedules and prefer quick access to food services. Traditional methods of ordering food often involve waiting in long queues or making phone calls, which can be time-consuming. An online system allows customers to browse restaurant menus and place orders from anywhere using the internet. It also helps restaurants manage customer orders in a more organized way. The system reduces human errors that may occur during manual order processing. Customers can easily select food items, view prices, and track their orders online. Digital payment options make the ordering process faster and more secure. Restaurants can also expand their business by reaching more customers through an online platform. Therefore, this system improves both customer convenience and restaurant service efficiency .

1.2 Problem Definition

Existing online food ordering systems often lack proper automation and efficient management features. Many systems still depend on manual processes, which can cause delays in order processing. This leads to incorrect orders, longer waiting times, and reduced customer satisfaction. Customers may face difficulties while browsing menus, placing orders, or tracking their deliveries. Restaurants also struggle to manage multiple orders efficiently during peak hours. The absence of real-time updates makes it difficult for administrators to monitor and control operations. Poor system integration can result in data errors and

miscommunication between users and restaurants. In addition, limited user interface design can negatively affect the overall user experience. These challenges highlight the need for a more reliable and automated online food ordering system. Therefore, an improved e-commerce based system is required to enhance efficiency, accuracy, and customer satisfaction.

1.3 Objective of Project

The main objective of the Online Food Ordering System is to develop an efficient and user-friendly e-commerce platform that allows customers to order food online from different restaurants. The system aims to simplify the traditional food ordering process by providing a digital interface where users can browse menus, select food items, and place orders easily. Another objective is to reduce manual work and improve accuracy in order management. The system also focuses on providing secure user authentication and reliable data management using a database. It helps restaurants manage orders, menus, and customer information efficiently. The project aims to enhance customer satisfaction by offering quick access to food services and real-time order updates. Additionally, it supports the integration of modern web technologies to improve system performance. Overall, the objective is to create a reliable and efficient online platform for food ordering using e-commerce technology.

1.4 Limitations of Project

The system may still depend on manual intervention for certain operations, which can reduce overall efficiency. Managing menus and handling multiple orders can become difficult if the system is not fully automated. In addition, the system may not provide complete real-time order tracking features, which can affect the user experience and order monitoring process.

1.4.1 Dependency of Internet Connectivity

The Online Food Ordering System requires a stable internet connection for users to access the platform and place orders. If the internet connection is slow or unavailable, users may face difficulties in browsing menus, selecting food items, and completing the ordering process. This dependency on internet connectivity can affect the overall accessibility and usability of the system.

1.4.2 Restaurent and menu Management

The system may support only a limited number of restaurants and menu items based on the database structure and system design. Managing a large number of restaurants, menus, and orders can become difficult if the system is not optimized for large-scale operations. This limitation may affect the scalability and efficiency of the platform when handling extensive data.

2. LITERATURE SURVEY

2.1 Introduction

The rapid growth of internet technology and e-commerce has significantly changed the way people order food from restaurants. Online food ordering systems have become popular because they provide customers with a convenient and efficient method to order food through web or mobile applications. Several studies have focused on developing systems that reduce manual work and improve order management in restaurants. Early research on Android-based food ordering systems introduced user authentication and order processing mechanisms to reduce manual errors and improve order accuracy.

Web-based customizable food ordering systems further improved the process by introducing features such as shopping carts and secure checkout methods. These systems enhanced the user experience by allowing customers to easily browse menus and place orders quickly. Later studies proposed digital dining systems that use centralized databases and dynamic menu management techniques to improve operational efficiency in restaurants. These systems help restaurant staff manage orders and menu updates more effectively while reducing human intervention.

Other research introduced integrated systems that combine digital ordering, kitchen order ticket (KOT) generation, and customer relationship management (CRM). These systems improve coordination between restaurant staff, kitchen staff, and administrators. Recent studies have also focused on secure and real-time online ordering systems using encryptionbased transaction mechanisms and recommendation algorithms. These advancements help improve customer satisfaction, enhance data security, and provide better administrative control. Overall, the literature survey shows that modern online food ordering systems aim to improve efficiency, accuracy, and user convenience in the food service industry.

2.2 Existing System

Traditional food ordering systems mainly rely on manual methods such as phone calls or direct ordering at restaurants. In these systems, customers either visit the restaurant or place orders through telephone communication. This process can be time-consuming and often leads to errors in order taking due to miscommunication between customers and restaurant staff. Managing multiple orders manually can also create confusion, especially during peak hours.

In many cases, restaurant staff must write down orders and process them manually, which increases the chances of mistakes and delays. Customers usually do not receive realtime updates about their orders, making it difficult to know the preparation or delivery status. Additionally, managing menus, updating food items, and maintaining customer records becomes difficult without a digital system.

Due to these limitations, traditional food ordering systems lack efficiency, accuracy, and proper order management. These challenges highlight the need for a more advanced and automated online food ordering system that can improve customer convenience and restaurant operations.



Fig 2.2: Existing system

2.3 Proposed System

The proposed system is a modern Online Food Ordering Platform (QuickBite) designed as AI-based meal planning, real-time nutrition tracking to provide users with a seamless, fast, and intelligent food ordering experience. The system integrates advanced features such as, and personalized recommendations to enhance user satisfaction and promote healthy eating habits. Users can easily browse food items through categorized menus, search for specific dishes, and view detailed information including price, ingredients, and nutritional values.

The platform supports user authentication, allowing customers to register, log in, and manage their profiles. It also connects users with verified local home chefs, ensuring authentic, high-quality homemade food while supporting small-scale businesses. The system includes a Smart Meal Planner that suggests meals based on user-defined calorie targets and dietary preferences, making it suitable for fitness-conscious individuals.

Additionally, the system provides a cart and order management module, enabling users to add items, place orders, and track their purchases efficiently. The interface is designed with a responsive and user-friendly UI/UX, ensuring smooth navigation across devices. Budget-friendly options and clear pricing make the platform accessible to a wide range of users. Overall, the proposed system aims to combine convenience, technology, and health awareness into a single platform, improving the traditional food ordering process by making it smarter, faster, and more personalized.

3.SYSTEM ANALYSIS

3.1 Functional Requirements

Functional requirements are the following

- The system should allow users to register and log in securely.
- The system should display a product or food catalog for browsing and selection.
- The system should allow users to add items to the shopping cart and place orders.
- The system should support secure online payment processing.
- The system should provide real-time order status tracking for users.
- The system should allow the administrator to manage products and monitor orders.

3.2 Non-Functional Requirements

3.2.1 Security

The system should ensure the protection of user data and transaction information from unauthorized access. Proper authentication and authorization mechanisms should be implemented to allow only registered users to access the system. Secure payment processing and data encryption techniques should also be used to maintain confidentiality and safety of sensitive information.

3.2.2 Performance

The system should provide fast response times for user requests such as browsing menus, adding items to the cart, and placing orders. It should be capable of handling multiple users simultaneously without significant delays. Efficient processing and optimized database management should ensure smooth system operation and quick order processing.

3.2.3 Usability

The system should provide a simple and user-friendly interface that allows users to easily browse menus, select food items, and place orders without difficulty. The design should be clear and easy to understand so that both customers and administrators can navigate the platform efficiently. Proper layout and responsive design should improve the overall user experience.

3.2.4 Reliability

The system should operate consistently and ensure that all functions work correctly without frequent failures.

3.3 Software Requirements

1. Operating system: Windows 10 or later
2. Web browser: Google Chrome / Microsoft Edge / Mozilla Firefox
3. Software: Visual Studio
4. Programming language: HTML, CSS, JavaScript, Python
5. Database: MySQL

3.4 Hardware Requirements

Processor: Intel Core i3 or higher

RAM: Minimum 4 GB

Hard Disk: 100 GB free space

3.5 Block Diagram of the proposed System

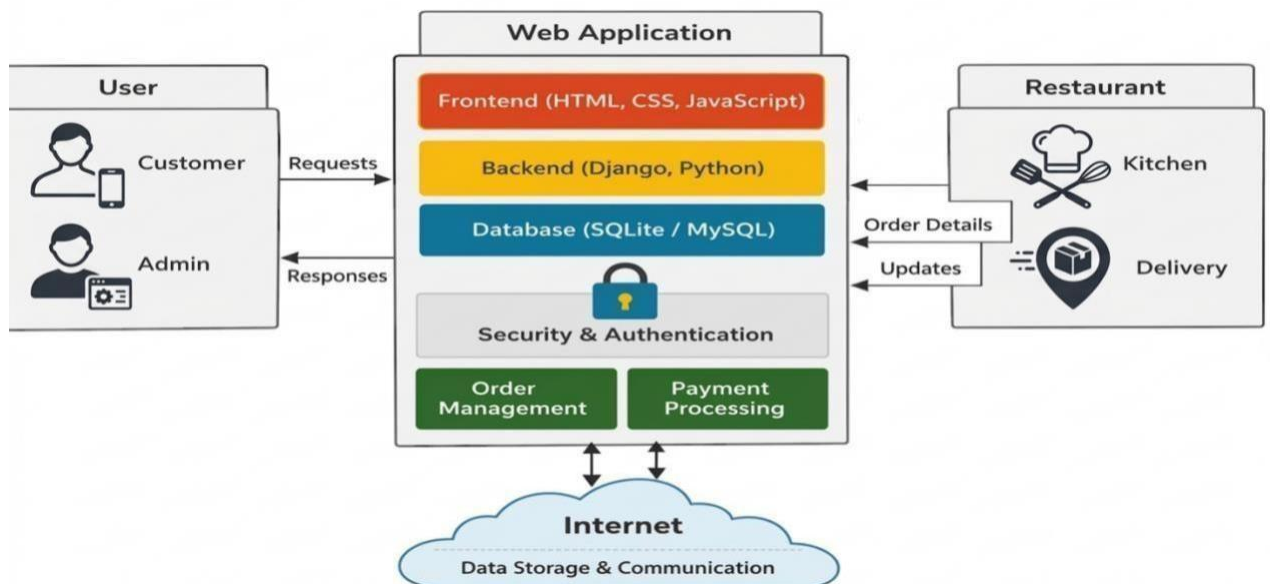


Fig 3.5 :Block diagram of the system

The block diagram of the proposed Online Food Ordering System illustrates the interaction between users, the web application, and the restaurant services. The system mainly consists of three major components: the User, the Web Application, and the Restaurant module. These components work together through the internet to provide an efficient online food ordering process.

The user module includes customers and administrators who interact with the system through a web interface. Customers can browse the food menu, place orders, and make online payments using the platform. Administrators are responsible for managing food items, monitoring orders, and updating system information. The user sends requests to the web application and receives responses such as order confirmation and order status updates.

The web application acts as the central part of the system. It consists of the frontend, backend, and database components. The frontend is developed using technologies such as HTML, CSS, and JavaScript, which provide an interactive user interface for customers and administrators. The backend is implemented using Django and Python, which handle the business logic, user authentication, and order processing. The database, such as SQLite or MySQL, stores important information including user details, menu items, and order records.

The system also includes security and authentication mechanisms to ensure safe access to the platform and protect user data. Order management and payment processing modules help manage customer orders and handle online transactions efficiently.

On the restaurant side, the system communicates with the kitchen and delivery modules. The kitchen receives order details from the system and prepares the food accordingly. After preparation, the delivery service handles the delivery process and updates the order status in the system. These updates are sent back to the user through the web application.

Overall, the block diagram represents the flow of information and communication between users, the application server, and restaurant services. This structure helps automate the food ordering process, improve efficiency, and provide a better user experience.

4. SYSTEM DESIGN

4.1 Introduction

System design is an important phase in the development of a software system where the overall structure and architecture of the system are planned. It defines how different components of the system interact with each other to achieve the desired functionality. The main purpose of system design is to convert system requirements into a structured design that guides the development process.

In the Online Food Ordering System, system design focuses on organizing the various modules such as user interface, database management, order processing, and payment handling. It describes how users interact with the system and how the system processes and stores information. Proper system design ensures that the system is efficient, reliable, and easy to maintain.

The design phase also helps developers identify the required technologies, system components, and data flow within the application. By creating diagrams such as block diagrams, use case diagrams, and system architecture diagrams, developers can clearly understand how the system will function. Overall, system design provides a blueprint for building a well-structured and scalable online food ordering platform.

4.2 UML Diagrams

Unified Modeling Language (UML) diagrams are used to visually represent the structure and behavior of a software system. They help developers understand how different components interact within the system. UML diagrams provide a clear representation of system functionality, architecture, and relationships between different modules. In the Online Food Ordering System, UML diagrams are used to model the system requirements and design before implementation.

These diagrams help identify the actors involved in the system and the processes they perform. They also show the interaction between users and different components of the system. UML diagrams make it easier for developers and stakeholders to understand the system design. The common UML diagrams used in this project include Use Case Diagram, Class Diagram, Activity Diagram, and Sequence Diagram. Each diagram represents a different aspect of the system structure and behavior. Overall, UML diagrams help in designing an efficient and well-organized online food ordering system.

4.2.1 Class Diagram

The UML diagram represents the structure of the Online Food Ordering System and shows the relationships between different classes in the system. The main classes included in the system are User, Customer, Admin, Restaurant, MenuItem, Order, Cart, Payment, and Delivery. The User class contains basic attributes such as userId, username, and password with functions like login, register, and logout. Customer and Admin are derived from the User class, where the Customer browses restaurants and places orders, and the Admin manages restaurants and orders.

The Restaurant class stores information such as restaurantId, name, and location and provides menu details. The MenuItem class contains information about food items including itemId, itemName, and price. The Order class manages order details such as orderId, userId, restaurantId, and orderStatus. The Cart class allows customers to add or remove food items before placing an order. The Payment class handles payment information including paymentId, amount, and payment method. It supports payment types such as CreditCardPayment and PayPalPayment. The Delivery class manages delivery details including deliveryId, deliveryPerson, and deliveryStatus, and updates the delivery process.

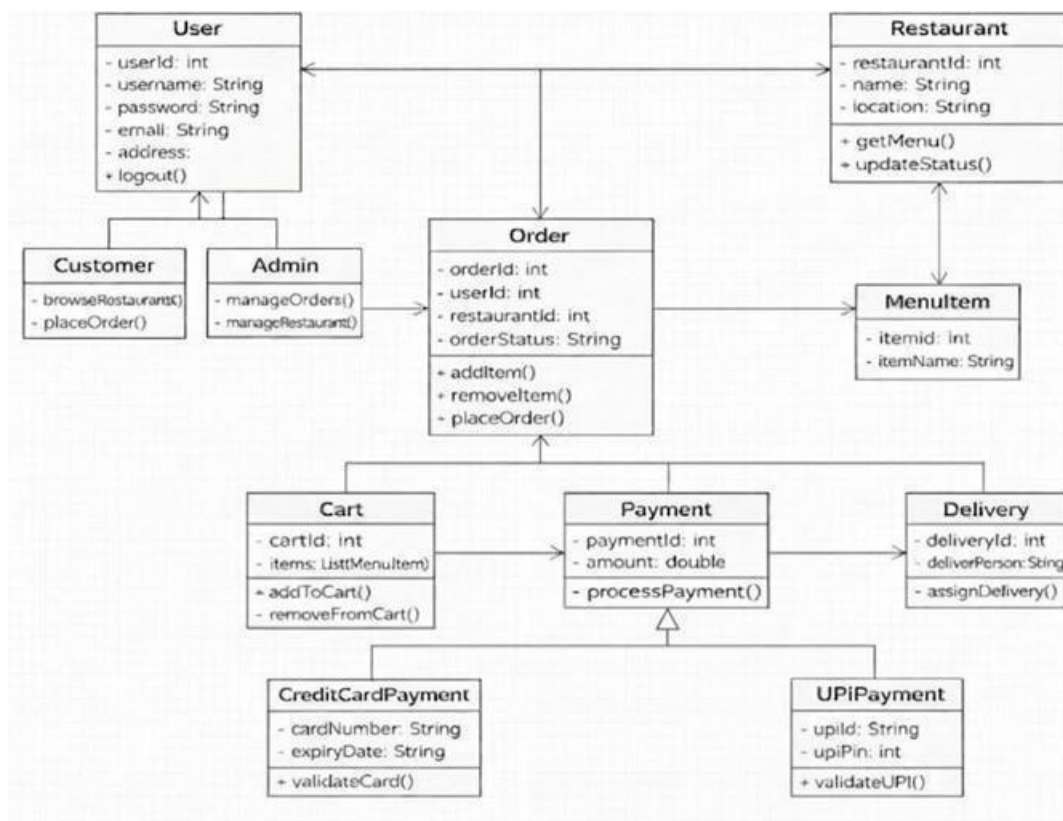


Fig 4.2.1 Class diagram

4.2.2 Activity Diagram

The activity diagram illustrates the workflow of the Online Food Ordering System in an e-commerce environment. The process begins when the user visits the website to access the food ordering platform. After accessing the website, the system checks whether the user wants to login or register before continuing with the ordering process. If the user chooses not to log in immediately, they can browse the available restaurants and view different food items offered by the restaurants. After browsing, the user selects the desired food items and adds them to the shopping cart. The system then allows the user to view the cart in order to review the selected items before proceeding further. Once the user confirms the selected items, the system provides an option to proceed to checkout for order confirmation. During the checkout process, the user makes the payment using the available payment methods.

After the payment is successfully completed, the system generates an order confirmation message for the user. Finally, the user can track the delivery status of the placed order until the food is delivered. This activity diagram helps to clearly represent the sequence of actions and decision points involved in the online food ordering process.

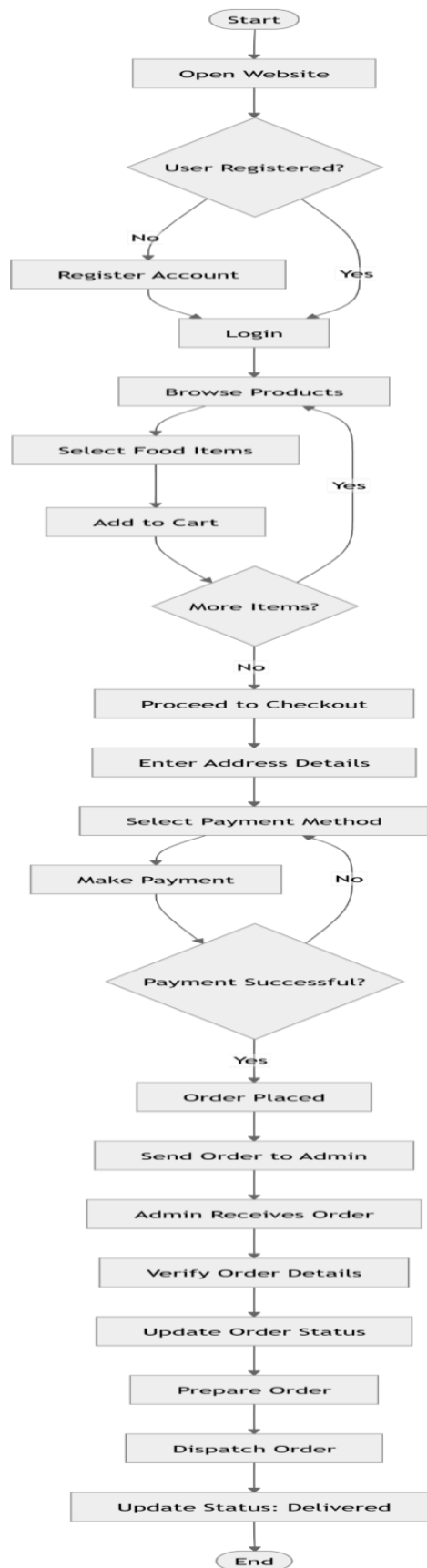


Fig 4.2.2:Activity diagram

4.2.3 Use case Diagram

The use case diagram represents the interaction between users and the Online Food Ordering System in an e-commerce environment. The main actors involved in the system are the Customer and the Admin. The Customer interacts with the system by registering and logging in to access the platform. After successful login, the customer can browse restaurants, view menus, select food items, add them to the cart, place orders, make payments, and track the delivery status.

On the other hand, the Admin is responsible for managing system operations. The Admin manages restaurant details by adding, updating, or removing them and also monitors customer orders and overall system activities.

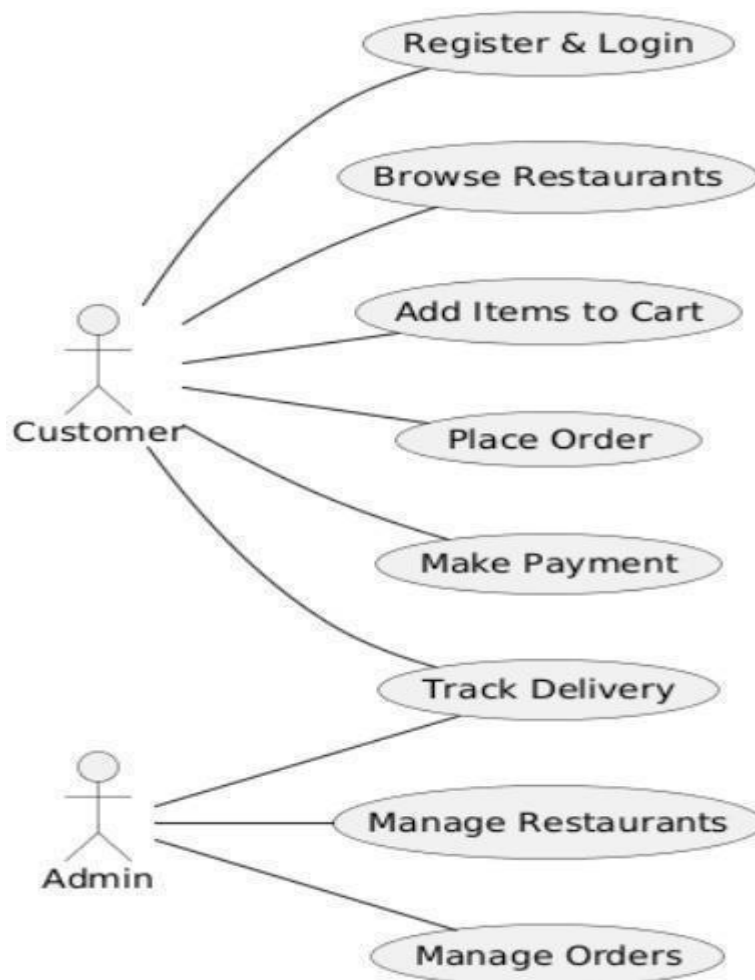


Fig 4.2.3:Usecase diagram

4.2.4 Sequence Diagram

The sequence diagram represents the interaction between different components of the Online Food Ordering System in an e-commerce environment. The main participants in the system include the Customer, Food Ordering System, Restaurant, and Payment Gateway. The process begins when the customer logs into the system or registers to access the platform. After successful login, the customer browses the available restaurants and views the menu items.

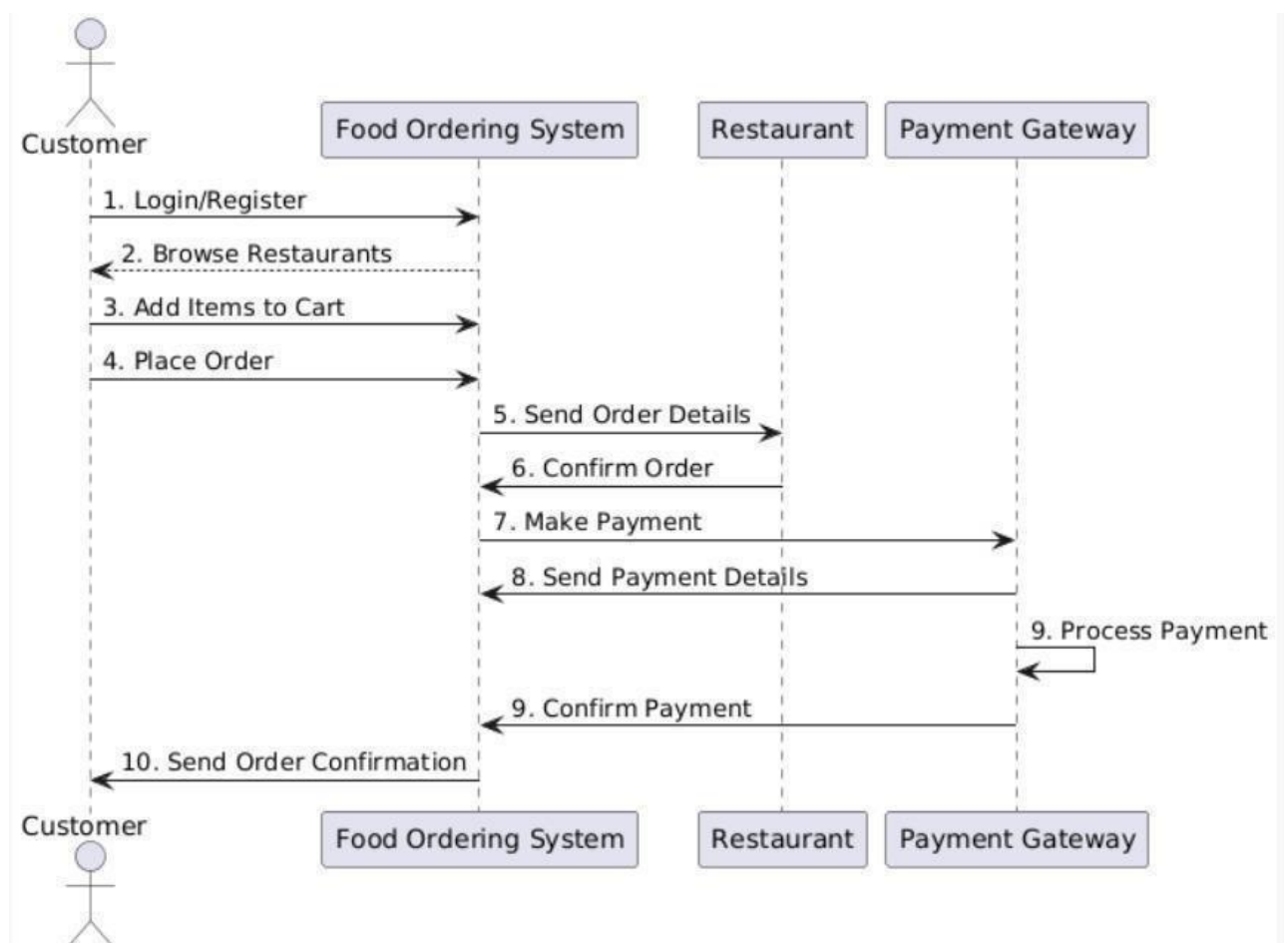


Fig 4.2.4:Sequence diagram

4.3 Modules

- [1] **User Management Module**-This module handles user registration, login, and authentication. It allows customers to create accounts and securely access the system.
- [2] **Restaurant Browsing Module**-This module allows users to view available restaurants and explore different food menus provided by each restaurant.
- [3] **Cart Management Module**-This module enables customers to add food items to the cart, update quantities, and remove items before placing an order.
- [4] **Order Management Module**-This module manages the order placement process and maintains order details such as selected items and order status.
- [5] **Payment Processing Module**-This module handles online payment transactions using available payment methods and ensures secure payment processing.
- [6] **Delivery Tracking Module**-This module allows customers to track the status of their food delivery after placing an order.
- [7] **Admin Management Module**-This module allows the administrator to manage restaurants, update menu items, and monitor customer orders.

5. IMPLEMENTATION AND RESULTS

5.1 Introduction

The implementation phase of the Online Food Ordering System focuses on developing the system based on the designed architecture and system requirements. In this phase, the different modules such as user registration, restaurant browsing, cart management, order processing, and payment handling are developed and integrated. The system is implemented using web technologies that allow users to interact with the platform through a user-friendly interface. Proper database connectivity is established to store and manage user data, menu details, and order information.

The result of the implementation is a fully functional online food ordering platform that allows customers to easily browse restaurants, select food items, and place orders through the system. The implemented system also supports secure user authentication and efficient order management. Customers can make online payments and receive order confirmations through the platform. Additionally, the system enables order tracking and delivery updates for users. Overall, the implementation and results demonstrate that the system successfully automates the food ordering process and improves convenience for both customers and administrators.

5.2 Method of Implementation

The Online Food Ordering System is implemented using a web-based architecture that integrates frontend, backend, and database components. The development process begins with designing the user interface using HTML, CSS, and JavaScript to provide an interactive and user-friendly environment for customers and administrators. These technologies help in creating web pages that allow users to browse restaurants, view menus, and place orders easily.

The backend of the system is developed using the Django framework with Python, which handles the application logic and communication between the user interface and the database. Django manages functions such as user authentication, order processing, and data management. It ensures that user requests are processed correctly and the required information is retrieved or stored in the database.

The database, such as SQLite or MySQL, is used to store important information including user details, restaurant data, menu items, and order records. The system processes customer orders, manages cart details, and handles payment information through the backend server. After successful order placement, the system updates the order status and provides confirmation to the user.

5.2.1 Source Code

#ADMIN

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>QuickBite | Admin Dashboard</title>

  <!-- External CSS -->

  <link rel="stylesheet" href="style.css">

  <!-- FontAwesome -->

  <link rel="stylesheet"

    href="https://cdnjs.cloudflare.com/ajax/libs/fontawesome/6.4.0/css/all.min.css">

  <style>

    body {

      margin: 0;

      font-family: Arial;

    }

    .admin-sidebar {

      width: 260px;

      background: #111;

      color: #fff;

      height: 100vh;

      position: fixed;

      padding: 20px;

    }

    .admin-content {

      margin-left: 260px;

      padding: 20px;

      background: #f4f7fa;

      min-height: 100vh
```

```
}
```

```
.nav-item {  
    padding: 10px;  
    cursor: pointer;  
}  
  
.nav-item.active {  
    background: orange;  
}  
  
.tab-content {  
    display: none;  
}  
  
.tab-content.active {  
    display: block;  
}  
  
table {  
    width: 100%;  
    border-collapse: collapse;  
}  
  
th, td {  
    padding: 10px;  
    border-bottom: 1px solid #ccc;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<!-- Sidebar -->
```

```
<div class="admin-sidebar">
```

```
<h2>QuickBite Admin</h2>
```

```

<div class="nav-item active" data-tab="dashboard">Dashboard</div>
<div class="nav-item" data-tab="orders">Orders</div>
<div class="nav-item" data-tab="products">Products</div>
<div class="nav-item" data-tab="users">Users</div>
<div class="nav-item" onclick="logout()">Logout</div>
</div>

```

```

<!-- Content -->

```

```

<div class="admin-content">

```

```

  <!-- Dashboard -->

```

```

  <div id="dashboard" class="tab-content active">
    <h2>Overview</h2>
    <p>Total Sales: <span id="sales"></span></p>
    <p>Total Orders: <span id="ordersCount"></span></p>
    <p>Total Users: <span id="usersCount"></span></p>
    <p>Total Products: <span id="productsCount"></span></p>
  </div>

```

```

  <!-- Orders -->

```

```

  <div id="orders" class="tab-content">
    <h2>Orders</h2>
    <table>
      <thead>
        <tr>
          <th>ID</th>
          <th>Name</th>
          <th>Total</th>
          <th>Status</th>
        </tr>
      </thead>
      <tbody id="ordersTable"></tbody>
    </table>

```

```

</div>

<!-- Products -->
<div id="products" class="tab-content">
  <h2>Products</h2>

  <table>

    <tbody id="productsTable"></tbody>

  </table>
</div>

<!-- Users -->
<div id="users" class="tab-content">
  <h2>Users</h2>

  <table>

    <tbody id="usersTable"></tbody>

  </table>
</div>
</div>
<script src="app.js"></script>
</body>
</html>

// Dummy Database (LocalStorage simulation)
const Database = {
  get: (key) => JSON.parse(localStorage.getItem(key)) || [],
  set: (key, data) => localStorage.setItem(key, JSON.stringify(data)),
  getCurrentUser: () => JSON.parse(localStorage.getItem("currentUser"))
};

// Sample Data (Run once)
if (!localStorage.getItem("orders")) {
  Database.set("orders", [
    { id: "101", name: "Pavan", totalAmount: 200, status: "Processing" }
  ]);
}

```

```

Database.set("products", [
  { id: 1, name: "Burger", price: 100, stock: 10 }
]);

Database.set("users", [
  { id: 1, name: "Admin", email: "admin@gmail.com", role: "admin" }
]);

localStorage.setItem("currentUser", JSON.stringify({ role: "admin" }));
}

// Authentication
const user = Database.getCurrentUser();
if (!user || user.role !== "admin") {
  alert("Access Denied");
}

// Tab Switching
document.querySelectorAll(".nav-item[data-tab]").forEach(item => {
  item.addEventListener("click", () => {
    document.querySelectorAll(".tab-content").forEach(t => t.classList.remove("active"));
    document.getElementById(item.dataset.tab).classList.add("active");
    loadData();
  });
});

// Load Data
function loadData() {
  const orders = Database.get("orders");
  const products = Database.get("products");
  const users = Database.get("users");

  // Stats
  document.getElementById("sales").innerText =
    orders.reduce((sum, o) => sum + o.totalAmount, 0);
  document.getElementById("ordersCount").innerText = orders.length;

```

```
document.getElementById("usersCount").innerText = users.length;
document.getElementById("productsCount").innerText = products.length;
```

// Orders Table

```
document.getElementById("ordersTable").innerHTML =
  orders.map(o => `
    <tr>
      <td>${o.id}</td>
      <td>${o.name}</td>
      <td>${o.totalAmount}</td>
      <td>${o.status}</td>
    </tr>
  `).join("");
```

// Products Table

```
document.getElementById("productsTable").innerHTML =
  products.map(p => `
    <tr>
      <td>${p.name}</td>
      <td>${p.price}</td>
      <td>${p.stock}</td>
    </tr>
  `).join("");
```

// Users Table

```
document.getElementById("usersTable").innerHTML =
  users.map(u => `
    <tr>
      <td>${u.name}</td>
      <td>${u.email}</td>
      <td>${u.role}</td>
    </tr>
  `).join("");
```

```

}

// Logout
function logout() {
    localStorage.removeItem("currentUser");
    alert("Logged out");
}

// Initial Load
loadData();

#DJANGO
#!/usr/bin/env python
import os
import sys
def main():
    """Run Django administrative tasks"""
    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'core_sys.settings')
    try:
        from django.core.management import execute_from_command_line
    except ImportError:
        raise ImportError("Django not installed")
    execute_from_command_line(sys.argv)
if __name__ == '__main__':
    main()

```

5.3 Algorithms and Flowcharts

[1] **User Registration Algorithm:** This algorithm manages the process of creating a new user account in the system. When a user enters registration details such as username, email, and password, the system verifies whether the information is valid and checks if the user already exists in the database. If the user does not exist, the system stores the user information in the database and creates a new account. This algorithm ensures secure user authentication and prevents duplicate registrations.

[2] **Login Authentication Algorithm:** This algorithm verifies the identity of users when they attempt to access the system. The user enters login credentials such as username and password, and the system compares these details with the stored data in the database. If the credentials match, the user is granted access to the system; otherwise, an error message is displayed. This algorithm ensures that only authorized users can access the platform.

[3] **Order Placement Algorithm:** This algorithm manages the process of selecting and placing food orders. The user browses available restaurants, selects food items from the menu, and adds them to the cart. After reviewing the cart, the user confirms the order, and the system generates an order record in the database. This algorithm ensures that all selected items and order details are stored properly.

[4] **Payment Processing Algorithm:** This algorithm handles the payment transaction after an order is placed. The user selects a payment method and submits payment details. The system processes the payment through a secure payment gateway and verifies the transaction. If the payment is successful, the system updates the order status and sends confirmation to the user. This algorithm ensures secure and reliable payment processing within the system.

5.4 Control Flow of the Implementation

5.4.1 Importing Dependencies

[1] Python: Python is a high-level, general-purpose programming language used to develop the backend of the Online Food Ordering System. It provides simple syntax, readability, and supports rapid development of web applications.

[2] Django: Django is a Python-based web framework used to build and manage the backend of the application. It provides features such as URL routing, authentication, database interaction, and secure web development.

[3] HTML: HTML (HyperText Markup Language) is used to design the structure of web pages. It helps in creating forms, menus, and layouts that allow users to interact with the online food ordering system.

[4] CSS: CSS (Cascading Style Sheets) is used to style and format the web pages. It enhances the visual appearance of the website by controlling colors, fonts, layouts, and responsiveness.

[5] JavaScript: JavaScript is used to add dynamic behavior and interactivity to web pages. It helps in form validation, dynamic content updates, and improving the overall user experience.

[6] SQLite / MySQL: SQLite or MySQL is used as the database management system to store and manage system data such as user information, restaurant details, menu items, and order records.

5.5 Output Screens

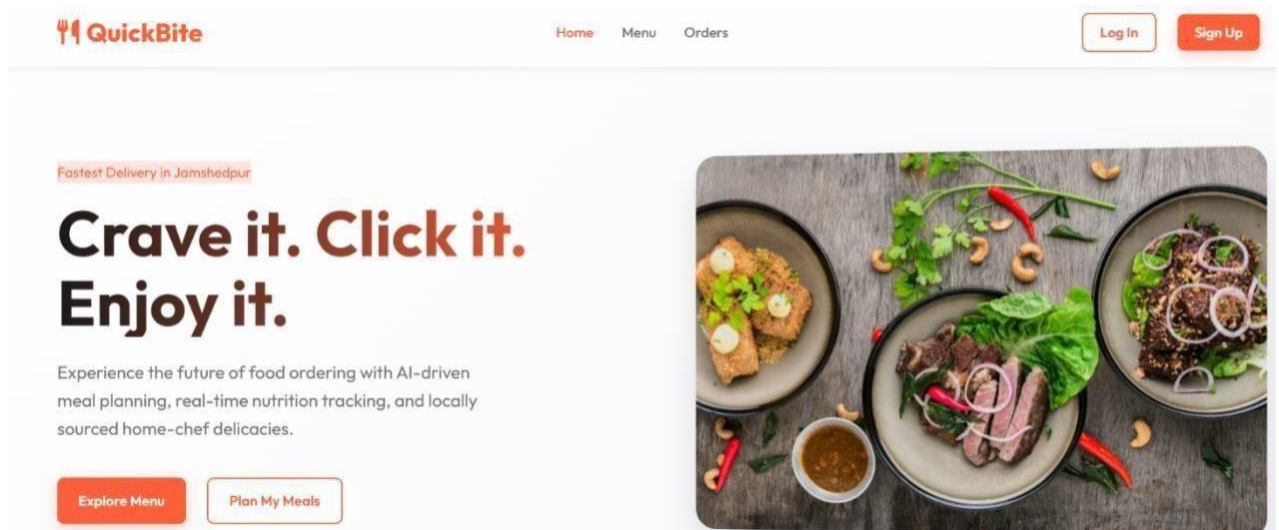


Fig 5.5.1: Output Screen– QuickBite Home Page UI

The screenshot represents the homepage of a food ordering web application named “QuickBite.” It is designed with a modern, user-friendly interface that focuses on simplicity, speed, and accessibility for users who want to order food online efficiently.

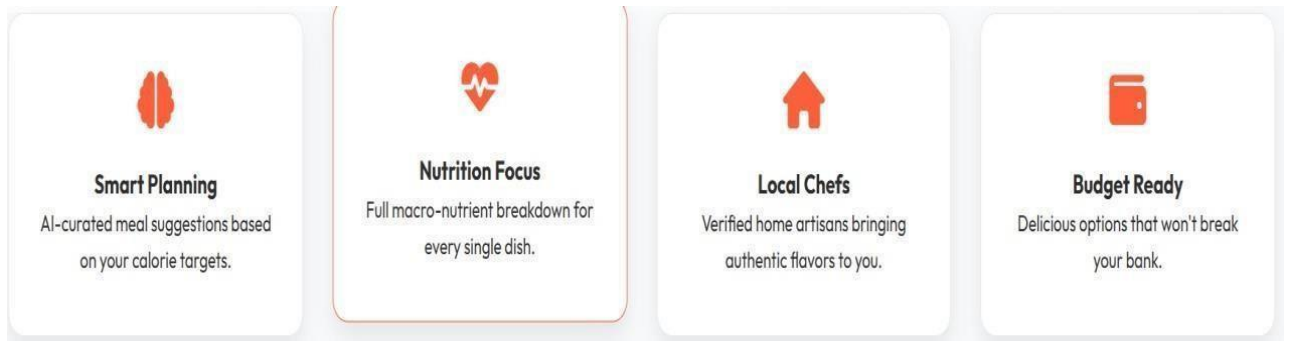


Fig 5.5.2 Output Screens– Features Section of QuickBite

The features section effectively communicates the core strengths of QuickBite, combining AI technology, health tracking, affordability, and local food services in a user-friendly interface.

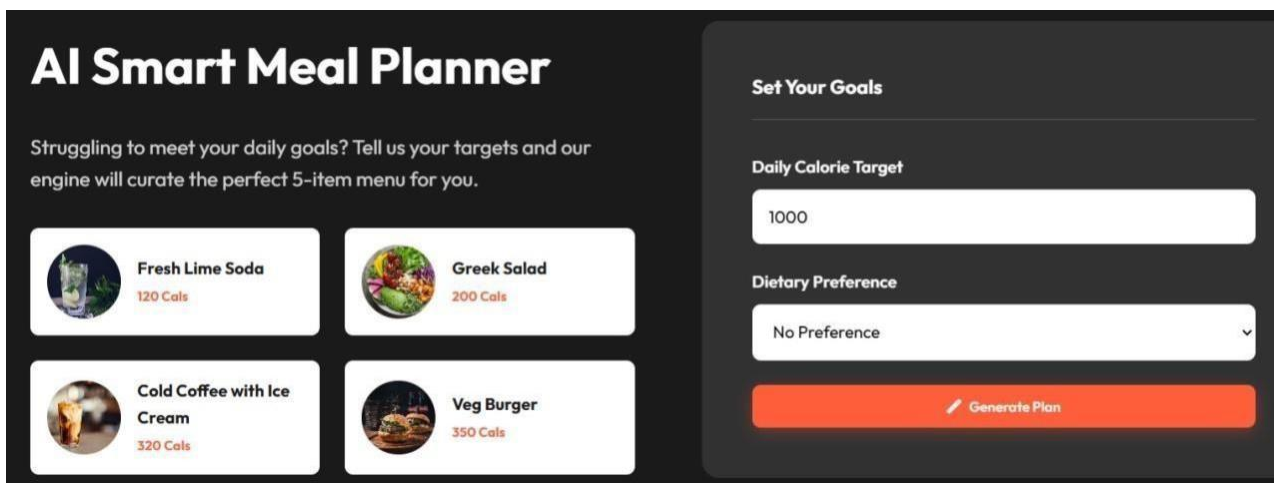


Fig 5.5.3: Output Screens – AI Smart Meal Planner

The AI Smart Meal Planner allows users to generate personalized meal plans based on their daily calorie targets and dietary preferences.

It uses an intelligent system to suggest suitable food items such as beverages, salads, and meals with their calorie values.

Users can input goals like calorie limits and select preferences to customize their diet plan. The interface displays recommended dishes in a structured card layout, improving usability and clarity.

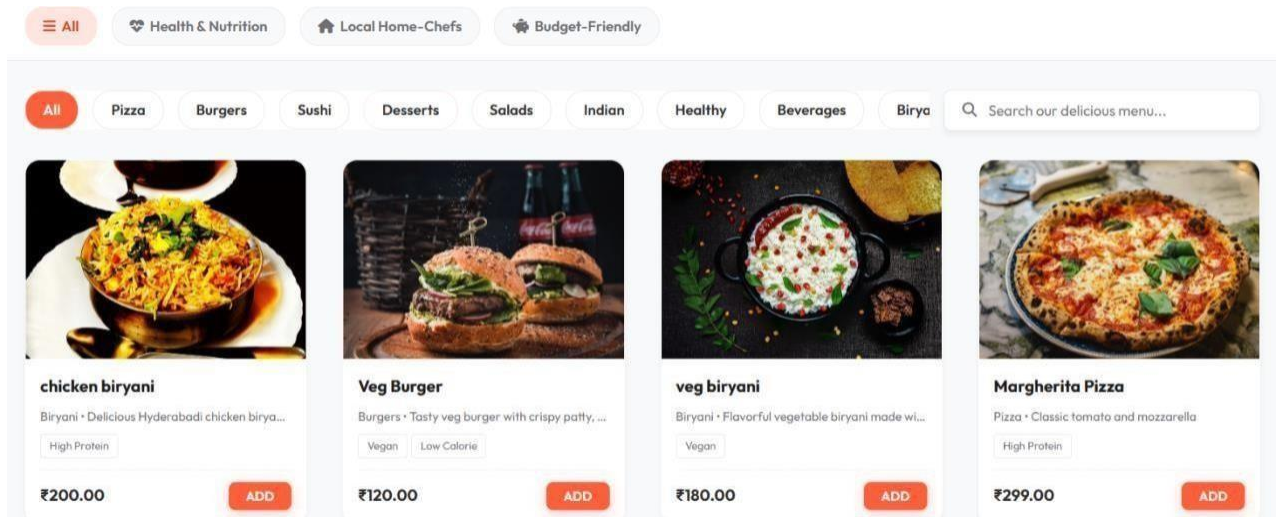


Fig 5.5.4: Output Screens – Menu Browsing Section

The menu browsing section allows users to explore various food items through categorized filters such as pizza, burgers, biryani, and more.

It provides a search bar for quick access to specific dishes, improving usability and navigation. Each food item is displayed in a card format with image, description, tags, price, and an “Add” button.

This layout helps users easily compare items and make quick ordering decisions.

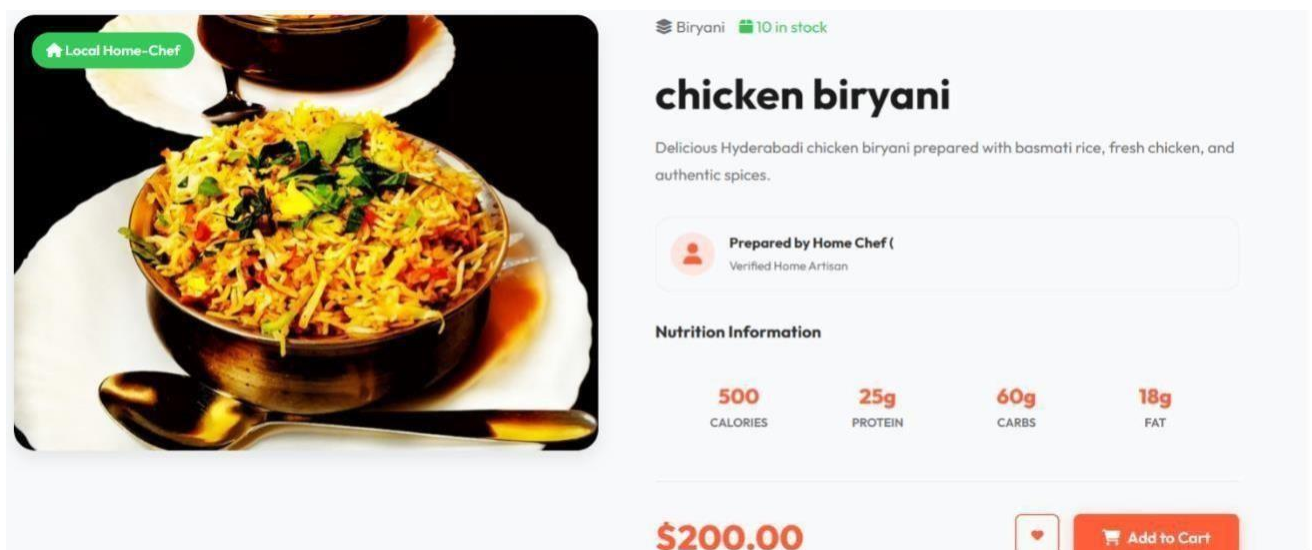


Fig 5.5.5: Output Screenshot – Food Item Details Page

The food item details page provides complete information about a selected dish, including image, description, and availability.

6. TESTING

6.1 Introduction to Testing

Testing is an important phase in software development that ensures the system works correctly and meets the specified requirements. It involves evaluating the functionality, performance, and reliability of the system to identify and fix errors or defects. The main purpose of testing is to ensure that the developed software operates as expected and provides accurate results.

In the Online Food Ordering System, testing is performed to verify that all modules such as user registration, login authentication, restaurant browsing, cart management, order processing, and payment handling function properly. Different testing methods are used to check the system's behavior under various conditions and to ensure that it provides correct outputs for given inputs. Testing also helps improve the quality, security, and performance of the application by detecting issues before the system is deployed. By conducting proper testing, developers can ensure that the system is reliable, user-friendly, and capable of handling user requests efficiently. Overall, testing plays a crucial role in delivering a stable and efficient online food ordering platform.

6.2 Types of Tests Consider

Unit Testing:Unit testing is used to test individual modules or components of the system separately. Each function, such as user registration, login authentication, menu browsing, and order placement, is tested independently to ensure that it performs the intended operation correctly.

Integration Testing:Integration testing is performed to verify the interaction between different modules of the system. It ensures that components such as the user interface, backend logic, and database work together properly when performing tasks like placing orders and processing payments.

System Testing:System testing is conducted to evaluate the complete system as a whole. It ensures that all system functionalities operate correctly together, including user authentication, restaurant browsing, cart management, order processing, and delivery tracking.

User Acceptance Testing (UAT):User Acceptance Testing is carried out to confirm that the system meets the requirements of end users. In this testing phase, users interact with the system to verify that it is easy to use, reliable, and capable of handling real-world scenarios effectively.

6.3 Various Test Case Scenarios Considered

TEST CASE ID	TEST CASE SCENARIO	INPUTS	EXPECTED OUTPUT	ACTUAL OUTPUT	STATUS
1	User Registration	Enter username,email, password	User account created successfully	As expected	Pass
2	User Login	Enter Valid username	User logged into the system	As expected	Pass
3	Browse Restaurents	Click on Browse Restaurents	List of available restaurents displayed	As expected	Pass
4	Make Payment	Enter Payment Details	Payment Processed Successfully	As expected	Pass
5	Place Order	Click Place Order	Order Created Successfully	As expected	Pass
6	Order Confirmation	After Successful Payment	Order Confirmation message displayed	As expected	Pass

Table 6.3: Test cases for proposed system

7.CONCLUSION AND FUTURE ENHANCEMENT

7.1 Project Conclusion

The Online Food Ordering System was developed to make the process of ordering food simple, fast, and convenient for users. The system allows customers to browse food items, add them to the cart, and place orders through an online platform. It reduces manual work and improves efficiency in managing food orders. The application provides a user-friendly interface for customers and an effective management system for administrators. It helps restaurants manage menus, orders, and customer details easily. The system also improves accuracy and reduces errors in the ordering process. Through this project, traditional food ordering is transformed into a digital system. It demonstrates the use of web technologies in the e-commerce environment. The project successfully meets its objectives and requirements. Overall, the system provides a reliable and efficient solution for online food ordering services.

7.2 Future Enhancement

The Online Food Ordering System can be further improved by adding advanced features to enhance functionality and user experience. In the future, the system can include real-time order tracking so that customers can monitor the status of their orders from preparation to delivery. Multiple secure payment options such as UPI, debit cards, credit cards, and digital wallets can be integrated to make the payment process more convenient. A rating and review system can also be added where customers can give feedback about food quality and delivery service. The system can be expanded by developing a mobile application for Android and iOS platforms to provide easier access for users. Location-based services can help customers find nearby restaurants quickly. Artificial Intelligence based recommendation systems can suggest food items based on customer preferences and previous orders. Promotional offers, discount coupons, and loyalty reward programs can also be implemented to attract more users. A delivery partner module can be developed to manage delivery activities and update delivery status. Advanced security measures can be added to protect user data and improve system reliability.

8. REFERENCES

8.1 Journals

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8.1 Books

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- [2] Web Development: HTML and CSS: Design and Build Websites, Jon Duckett
- [3] Database Management: Database System Concepts, Abraham Silberschatz, Henry F. Korth, and S. Sudarshan

8.2 Sites

- [1] <https://www.zomato.com>
- [2] <https://www.swiggy.com>
- [3] <https://www.dominos.co.in>